

Meshary Asdala Aquino

Software Engineer

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SUMMARY

Backend-leaning software engineer with 2 years shipping production systems where correctness matters: payroll compliance, encrypted data pipelines, and offline-first sync. Strongest in Python and TypeScript backends with PostgreSQL, with full-stack delivery in React and FastAPI.

EXPERIENCE

IT Specialist — M. Abubakar Construction and Engineering | Zamboanga City | June 2025 – June 2026

- Cut document and certificate reconciliation from three days to under four hours by replacing a shared-spreadsheet process with a normalized PostgreSQL tracking system, authenticated across three user tiers with role-based access control.
- Eliminated 15+ hours of weekly manual data entry across a 50-employee payroll cycle by building a Python desktop application (PyQt6) that automated statutory deductions, contribution computations, and payslip generation end to end.

Freelance Software Engineer (part-time, concurrent) | September 2024 – Present

- Built real-time video and messaging for LearnMate, a cross-platform tutoring app serving WMSU undergrad researchers, by designing the Firebase data model, role-based authentication, and WebRTC signaling layer. (*Dart, Flutter, Firebase*)
- Delivered on-device wood knot detection as sole engineer for AdZU undergrad researchers by training a YOLOv8 model on 12,000 images and shipping the full ONNX-to-TFLite inference pipeline. (*Python, TensorFlow Lite*)

Software Developer Intern — Regional Trial Court, Region 9 | Zamboanga City | August 2024 – April 2025

- Kept court operations running for 10+ daily staff through repeated network outages by building an offline-first desktop client with automated cloud synchronization.
- Reconciled 1,000+ legal records after outages with zero data loss by engineering bi-directional sync with conflict resolution between offline clients and the live database.

PROJECTS

Enterprise Payroll Web Application — *Next.js 14, React 18, TypeScript, FastAPI, PostgreSQL, TailwindCSS*

- Delivered BIR TRAIN Law-compliant payroll across semi-monthly, weekly and daily frequencies by building a tax engine implementing the 2023 revised schedules with 2025 SSS, PhilHealth and Pag-IBIG contribution tables.
- Secured employee and bank data at rest with AES-256 field encryption, JWT authentication with bcrypt hashing, RBAC on every protected REST endpoint, and an immutable audit trail.

Coupleledger — *Dual-layer household ledger (React 18, TypeScript, Vite, TailwindCSS)*

- Designed a dual-layer access model enforcing strict data isolation between two roles on a shared workspace — every write operation validated and audit-logged before commit, preventing cross-role data leakage.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Dart, SQL, PHP, HTML/CSS

Frontend: React 18, Next.js 14, TailwindCSS, Flutter, PyQt6, responsive UI, WebRTC

Backend & Data: FastAPI, PostgreSQL, MySQL, SQL Server, Firebase, REST APIs, relational schema design, JWT, RBAC

Tools: Git/GitHub, Docker, Vite, pytest, AWS, Linux, Agile/Scrum

EDUCATION

Mapúa University — Master of Engineering in Computer Engineering (in progress), 2026 – present

Western Mindanao State University — BS Computer Engineering, Zamboanga City, August 2021 – May 2025

GPA: 1.71 (1.00 = highest; first-class range)

CERTIFICATIONS

Civil Service Professional Eligibility (CSC) • Data Associate Pathway (DAP, 2025) • Networking Basics & Python Essentials 1 (Cisco, 2025)